

HACKMYSURU 1.0 — PHASE 1

CivicMysuru

Time-aware jurisdiction routing for citizen civic complaints

Chosen sub-problem: Routing

THE SITUATION

Mysuru's civic reality, right now

550-600 tonnes

of garbage generated every day

~7 lakh tonnes

backlog of untreated waste

150+ tonnes/day

construction waste, often dumped illegally at night

1 CMC + 4 TPs + 8 GPs

being actively absorbed into Mysuru City Corporation

CORE REALITIES

Why this is harder than it looks

People will lie

Citizens can report freely, even anonymously — which also invites fake complaints and spam.

Jurisdiction is genuinely unclear

Mysuru's boundaries are actively changing this year. A hardcoded map is already wrong.

Not every complaint is equally urgent

A pothole and a hazard on a school route can't be treated the same way.

Bad input is guaranteed

Duplicate reports, reused photos, and wrong locations will happen — the system must handle them, not assume they won't.

THE SOLUTION

What CivicMysuru actually does

01

Report

Citizen submits area, issue type, description, optional photo, and GPS location.

02

Classify

A rule-based classifier suggests the issue type from the description/filename.

03

Route

A time-aware jurisdiction engine resolves the correct authority as of today's date.

04

Verify

Priority, verification score, duplicate check, and reused-evidence detection all run automatically.

Boundaries change. The routing engine already knows.

Every area-to-authority mapping is a dated record, not a fixed fact.

Example: Hootagalli belonged to its own Town Panchayat until 1 April 2026 — from that date onward, the exact same area routes to Mysuru City Corporation. No code change, no redeploy.

`POST /api/admin/jurisdiction-change` → `simulates a real boundary merger live, on stage`

An unmatched or disputed area is never silently guessed — it's flagged for manual review instead.

Duplicates and fake evidence, actually caught

Duplicate detection

Two similar reports in the same area and issue category → the second is flagged duplicate: true.

Reused evidence

The same photo submitted on two different complaints → verification score drops and it's flagged as possible fake evidence.

Unmatched jurisdiction

An area with no matching record routes to manual review instead of a confident-sounding guess.

Routing is only half the job. Resolution is the other half.

A routed authority label means nothing unless the right staff can find their own queue.

01

Authority filter

The Dashboard's queue can be narrowed to one authority at a time — simulating "log in as Nanjangud City Municipal Council staff."

02

Isolated queue

Only complaints where authority matches the selected office are shown — Hootagalli's newly-absorbed queue looks different from Nanjangud's.

03

Resolve

Start Work / Mark Resolved call PATCH `/api/complaints/{id}/status` — the same status the citizen sees on their own tracker.

Honest scope: this is demo authentication, not enforced access control — the filter simulates role separation for the demo. A real deployment ties each staff account to exactly one authority server-side.

Tech stack

Frontend

- Static HTML / CSS / JS
- No build step, no framework overhead
- Pages: Home, Login, Report, Dashboard, AI Detection

Backend

- FastAPI + SQLAlchemy + SQLite
- Modular: routing, jurisdiction, classifier engines
- CORS-enabled REST API, auto-generated /docs

Deployment

- Frontend → Netlify (static)
- Backend → Render (Python web service)
- python main.py runs it locally, unchanged

Bonus feature (not our chosen sub-problem): a standalone AI Detection page calls the same rule-based classifier for instant, explainable photo classification — clearly scoped in our decision log as distinct from the brief's "Detection Without Reporting" sub-problem.

What we'd improve with more time

- Storage is SQLite, not a versioned GIS system — sufficient to prove the concept for this MVP.
- Fake-evidence detection compares filenames, not real image content (a placeholder for perceptual hashing once real image uploads exist).
- Duplicate detection uses word overlap within the same area, not GPS distance — the GPS button already captures coordinates for a future upgrade.
- The AI classifier is explainable and rule-based by design, not a trained vision model.

CivicMysuru

Report. Route. Resolve.

Chosen sub-problem

Routing

Core realities answered

4 of 4 (lying, jurisdiction, priority, bad input)

Live demo endpoints

11 REST API routes, auto-documented at /docs

Honest limitations disclosed

4, each with a stated upgrade path

Live demo, video walkthrough, and decision log linked in resource.md at the repository root.